

Auteur: Stan Van Campenhout
 Studierichting: Informatica
 Oefeningenreeks 3H nr. 8.7

$$r = 2 \cos 6\theta$$

$$\text{dom} = \mathbb{R}$$

$$p = \frac{\pi}{3}$$

$$\rightarrow \text{BD} = [0, \frac{\pi}{3}]$$

$$r = 0 \Leftrightarrow 2 \cos 6\theta = 0 \Leftrightarrow 6\theta = \frac{\pi}{2} + k\pi \Leftrightarrow \theta = \frac{\pi}{12} + k\frac{\pi}{6}, \forall k \in \mathbb{Z}$$

$$\rightarrow \left\{ \frac{\pi}{12}, \frac{\pi}{4} \right\}$$

$$r' = 0 \Leftrightarrow -2 \sin 6\theta = 0 \Leftrightarrow 6\theta = k\pi \Leftrightarrow \theta = k\frac{\pi}{6}, \forall k \in \mathbb{Z}$$

$$\rightarrow \left\{ 0, \frac{\pi}{6}, \frac{\pi}{3} \right\}$$

θ	0		$\frac{\pi}{12}$		$\frac{\pi}{6}$		$\frac{\pi}{4}$		$\frac{\pi}{3}$
r	+	+	0	-	-	-	0	+	+
r'	0	-	-	-	0	+	+	+	0
r	M	$\nearrow$	$\nearrow$	$\nearrow$	m	$\searrow$	$\searrow$	$\searrow$	M
	2				-2				2

3H-8.7-stan.van.campenhout.INF.png

References